

HVAC Controls & Building Management System Specifications

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Project: Lakefront Office Tower — Phase II

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SECTION 23 09 23 — DIRECT DIGITAL CONTROLS FOR HVAC

1.0 SCOPE

Furnish, install, program, and commission a fully integrated BMS providing DDC of all mechanical systems. BMS interfaces with electrical, fire alarm, security, and lighting via BACnet/IP protocol.

2.0 BMS NETWORK ARCHITECTURE

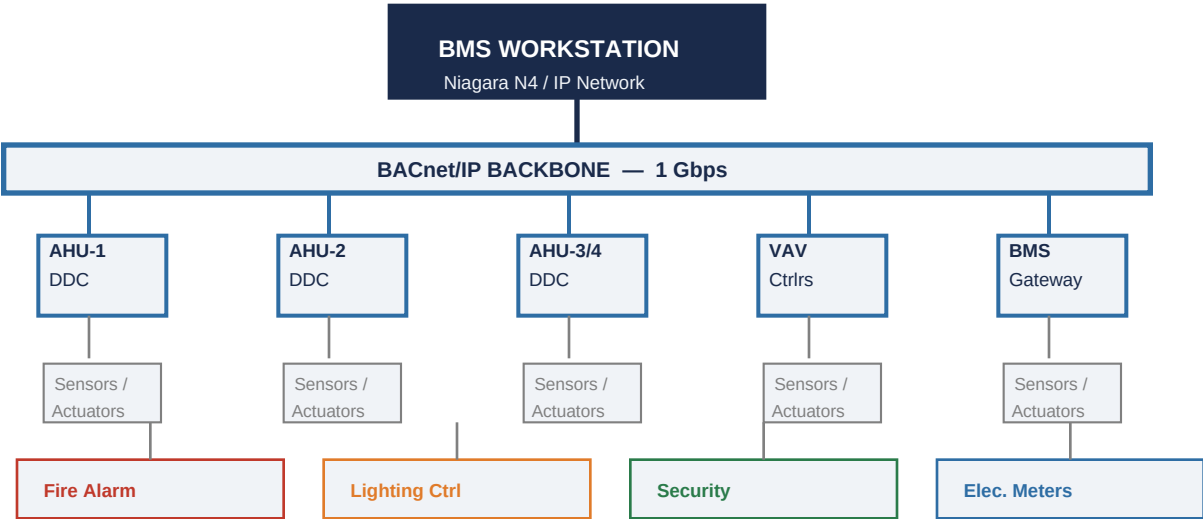


Figure 3 — BMS Network Architecture (Schematic — Not to Scale)

3.0 AHU CONTROL SEQUENCES

3.1 Supply Air Temperature Reset

OAT (°F)	SAT Setpoint (°F)
≥ 75	55
65	57
55	60
≤ 45	65

3.2 Duct Static Pressure Reset

Setpoint resets from 1.5 in. w.g. toward 0.5 in. w.g. when all VAV boxes satisfy load with dampers <90% open. Decrease 0.1 in./5-min interval. ASHRAE 90.1-2022 §6.5.3.2.

4.0 I/O POINTS LIST — AHU-1

Point ID	Description	Type	Range / State
AHU1-MAT	Mixed Air Temp	AI	–20 to 120°F

Point ID	Description	Type	Range / State
AHU1-SAT	Supply Air Temp	AI	40 to 120°F
AHU1-RAT	Return Air Temp	AI	40 to 120°F
AHU1-OAT	Outdoor Air Temp	AI	–30 to 130°F
AHU1-SAP	Duct Static Pressure	AI	0 to 5 in. w.g.
AHU1-CHWV	CHW Valve Position	AO	0–100%
AHU1-HHWV	HHW Valve Position	AO	0–100%
AHU1-OADMP	OA Damper	AO	0–100%
AHU1-SFVFD	Supply Fan VFD Speed	AO	0–100% (min 15%)
AHU1-SFSS	Supply Fan Start/Stop	DO	Off / On
AHU1-SFST	Supply Fan Status	DI	Off / On
AHU1-SFFL	Supply Fan VFD Fault	DI	Normal / Fault
AHU1-FLTST	Filter ΔP Status	DI	Normal / Dirty
AHU1-FZDTK	Freeze-Stat Alarm	DI	Normal / Trip

END OF SECTION 23 09 23